

ComputerVision HW2

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1 Task 1

I applied the image featurization functions taught in class which included find contours. I had trouble with warp image because I wanted to use the paper as a background to help calculate the book size but it was not reliable and gave the book an odd shape. I used find contours to calculate the height and width of paper sides and then use the ratio of pixels to cm from the paper calculation to determine book width and height. Finally, we calculated % error for width and height calculations of the book. I also used OOP to make it easier to find the size of a rectangle.

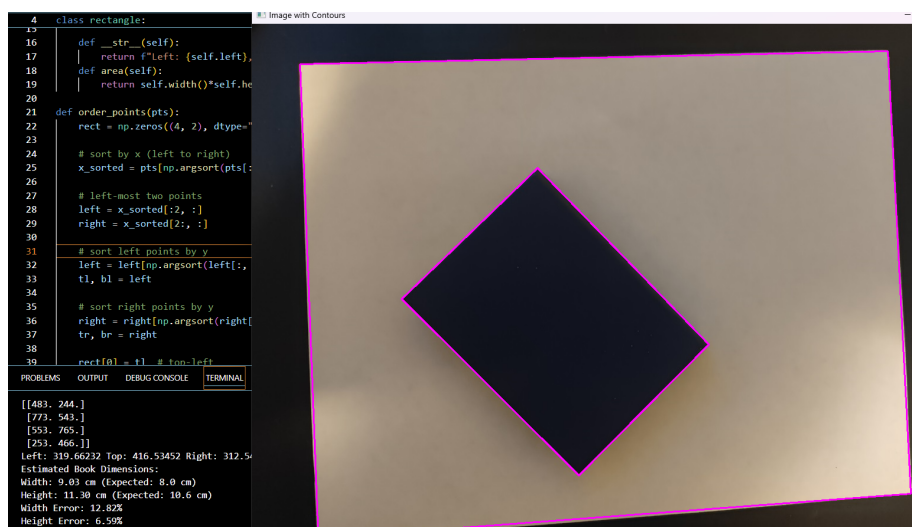


Figure 1: Successfully calculated book dimensions

2 Task 2

Again, my attempt to warp the image around the road sign was unsuccessful. My program would be stuck finding contours in random parts of the image

that had noise. If I could have warped only around the road signs maybe this program would be perfect. My percent error is pretty poor though so this algorithm should not be used by Tesla. I mainly resorted to image preprocessing techniques. I mainly applied blur along with image thresholding.

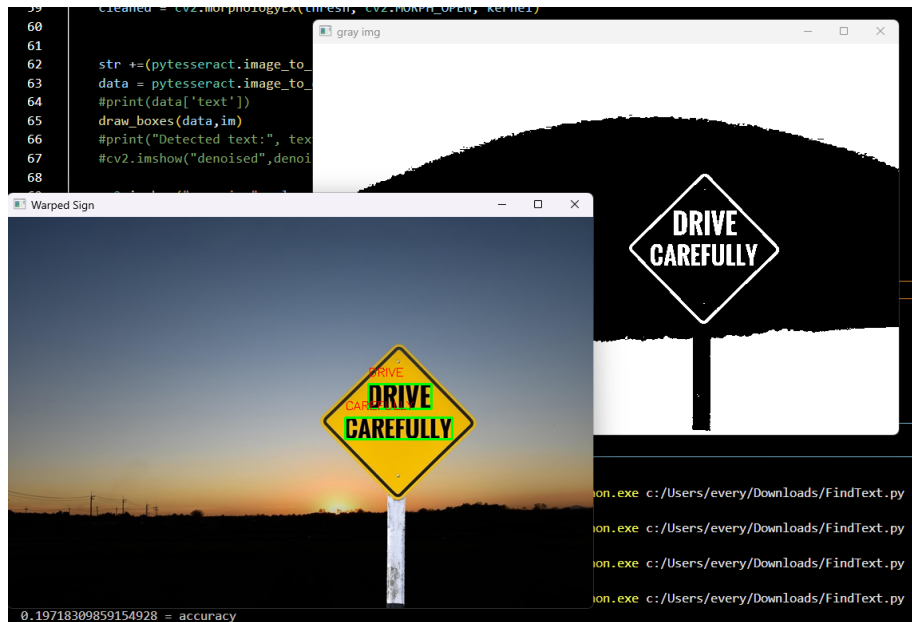


Figure 2: Successfully detected text